

D882

Secondary Biology Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

For a lab on the first day of the cell unit, a teacher places red onion skin cells in a 10% salt solution. The onions came from the school's garden, and the class uses the new digital microscopes. The salt solution has far more solute than the cell's cytoplasm. What should students see after five minutes?

- A. Salt moves into the cell until the cell is full of salt.
- B. The cell swells and bursts as water rushes in.
- C. Nothing changes, because the cell wall blocks all water movement.
- D. The cell membrane pulls away from the cell wall as water leaves the cell.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

2. [TRANSFER — reversed stem / different variable]

A red blood cell is placed in distilled water. Which term describes the water, and what happens to the cell?

- A. Isotonic; no change.
- B. Hypertonic; the cell shrinks.
- C. Hypotonic; the cell shrinks.
- D. Hypotonic; water enters and the cell may burst.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

3.

Students examine a case study of pancreatic cells that release large amounts of digestive enzymes, which are proteins. The case study was written for a tenth-grade audience and includes colorful diagrams. Which organelles would the students expect to be most abundant in these cells?

- A. Only the nucleus, because it contains the genes for the enzymes
- B. Rough endoplasmic reticulum and Golgi apparatus
- C. Smooth ER and lysosomes
- D. Chloroplasts and a central vacuole

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

4.

A human skin cell from a scraped knee divides to heal the wound. The student who scraped it is on the track team and runs every afternoon. A human body cell has 46 chromosomes. How many chromosomes does each new skin cell have?

- A. 92
- B. 23
- C. 46
- D. It depends on how fast the student heals.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In a seed company scenario used in class, two pea plants that are both heterozygous for purple flowers (Pp) are crossed. The company sells seeds in three states and ships in recyclable packaging. The cross produces 400 offspring. About how many will have white flowers?

- A. 300
- B. 200
- C. 100
- D. 133

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A purple-flowered plant of unknown genotype is crossed with a white-flowered plant (pp). The offspring are about half purple and half white. What is the genotype of the purple parent?

- A. pp
- B. PP
- C. It cannot be determined from a test cross.
- D. Pp

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher uses a dog-breeding scenario: black coat (B) is dominant to brown (b), and short hair (S) is dominant to long (s). Both parents are BbSs and are champion show dogs from the same kennel. What fraction of puppies is expected to be brown with long hair?

- A. 1/16
- B. 1/2
- C. 9/16
- D. 3/16

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A BbSs dog is crossed with a bbss dog. What fraction of the offspring is expected to be bbss?

- A. 1/16
- B. 1/2
- C. 1/4
- D. 0, because the dominant alleles hide the recessive ones.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A flower grower at a farmers market crosses red snapdragons with white snapdragons. The grower has been in business for 20 years and sells mostly to wedding planners. All the offspring are pink. Which inheritance pattern does this show?

- A. Complete dominance
- B. Incomplete dominance
- C. Sex-linked inheritance
- D. Codominance

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A pedigree used in class shows a mother with normal color vision whose father was color-blind; she marries a man with normal vision. Red-green color blindness is X-linked recessive. The family lives on a farm and has three dogs. What is the chance that a son of this couple will be color-blind?

- A.** 1/2
- B.** 1, because her father was color-blind.
- C.** 0, because the father has normal vision.
- D.** 1/4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A color-blind man and a woman with two normal X alleles have children. What is expected?

- A.** Half of the sons are color-blind.
- B.** All daughters are carriers and no sons are color-blind.
- C.** No children carry the allele.
- D.** All daughters are color-blind.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a lab using an enzyme from pineapple, students heat samples to 20, 40, 60, and 90 degrees C before adding gelatin. The pineapple was bought on sale and cut by the teacher before class. Activity rises from 20 to 40 degrees C. What result at 90 degrees C best fits enzyme behavior?

- A.** Very low activity, because the enzyme's shape was changed by heat.
- B.** Normal activity, because enzymes are not changed by the reactions they speed up.
- C.** Activity drops because the enzyme gets used up at 40 degrees C.
- D.** The highest activity, because heat always speeds up reactions.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

~~For a lab on the first day of the cell unit, a teacher places red onion skin cells in a 10% salt solution. The onions came from the school's garden, and the class uses the new digital microscopes.~~ The salt solution has far more solute than the cell's cytoplasm. What should students see after five minutes?

Target word: what happens to the cell **Decides:** The solution is hypertonic, so water leaves.

A	Confusion set	Osmosis is the movement of water; the membrane limits salt movement.
B	Inverted operation	That happens in a hypotonic solution, and plant cell walls usually prevent bursting.
C	Absolute language	The wall is permeable to water; it only resists swelling.
D	Correct	In a hypertonic solution water moves out; plant cells undergo plasmolysis.

2. Answer D Concept 1 • transfer

A red blood cell is placed in distilled water. Which term describes the water, and what happens to the cell?

Target word: tonicity and result **Decides:** Distilled water has less solute than the cell.

A	Adjacent category	Isotonic means equal solute levels, which is not true here.
B	Inverted operation	Reverses the relative solute levels.
C	Dropped element	Correct term, but drops the direction of water movement.
D	Correct	Water moves toward the higher solute concentration inside the cell.

3. Answer B Concept 2

Students examine a case study of pancreatic cells that release large amounts of digestive enzymes, which are proteins. ~~The case study was written for a tenth-grade audience and includes colorful diagrams.~~ Which organelles would the students expect to be most abundant in these cells?

Target word: organelles for protein secretion **Decides:** The cells make and ship out proteins.

A	Dropped element	Genes are needed, but the question asks what is abundant for making and shipping protein.
B	Correct	Ribosomes on rough ER make secreted proteins; the Golgi packages and ships them.
C	Adjacent category	Smooth ER makes lipids; lysosomes break down material, not secrete proteins.
D	Confusion set	These are plant organelles, not found in pancreatic cells.

4. Answer C Concept 3

A human skin cell from a scraped knee divides to heal the wound. ~~The student who scraped it is on the track team and runs every afternoon.~~ A human body cell has 46 chromosomes. How many chromosomes does each new skin cell have?

Target word: chromosome number after mitosis **Decides:** Skin healing uses mitosis, which keeps the number.

A	Inverted operation	Doubles instead of keeping the number; DNA is copied, then split.
B	Confusion set	That is the gamete number after meiosis.
C	Correct	Mitosis produces two cells identical to the parent.
D	Camouflage	Healing speed does not change chromosome number.

5. Answer C Concept 4

In a seed company scenario used in class, two pea plants that are both heterozygous for purple flowers (Pp) are crossed. ~~The company sells seeds in three states and ships in recyclable packaging.~~ The cross produces 400 offspring. About how many will have white flowers?

Target word: number of recessive phenotype **Decides:** Pp x Pp gives 1/4 pp.

A	Inverted operation	That is the purple count, the dominant phenotype.
B	Confusion set	Uses the 1/2 heterozygous fraction, a genotype, not white flowers.
C	Correct	1/4 of 400 = 100.
D	Wrong stem number	Treats 1:2 as thirds instead of using the 1:2:1 ratio.

6. Answer D Concept 4 • transfer

A purple-flowered plant of unknown genotype is crossed with a white-flowered plant (pp). The offspring are about half purple and half white. What is the genotype of the purple parent?

Target word: infer parent genotype **Decides:** A 1:1 ratio in a test cross means the parent was heterozygous.

A	Confusion set	A pp plant is white, not purple.
B	Inverted operation	PP x pp would give all purple offspring.
C	Absolute language	A test cross exists exactly to reveal this.
D	Correct	Pp x pp gives 1/2 Pp (purple) and 1/2 pp (white).

7. Answer A Concept 5

A teacher uses a dog-breeding scenario: black coat (B) is dominant to brown (b), and short hair (S) is dominant to long (s). Both parents are BbSs ~~and are champion show dogs from the same kennel.~~ What fraction of puppies is expected to be brown with long hair?

Target word: probability of double recessive **Decides:** Independent genes: multiply 1/4 x 1/4.

A	Correct	P(bb) x P(ss) = 1/4 x 1/4 = 1/16.
B	Inverted operation	Adds 1/4 + 1/4 instead of multiplying.
C	Confusion set	That is the fraction with both dominant traits.
D	Adjacent category	That is a one-dominant, one-recessive class.

8. Answer C Concept 5 • transfer

A BbSs dog is crossed with a bbss dog. What fraction of the offspring is expected to be bbss?

Target word: test-cross probability **Decides:** Each gene gives 1/2 recessive offspring.

A	Wrong stem number	Uses the fraction from a BbSs x BbSs cross.
B	Dropped element	Counts only one gene.
C	Correct	1/2 x 1/2 = 1/4.
D	Absolute language	The bbss parent gives only recessive alleles, so bbss offspring are possible.

9. Answer B Concept 6

A flower grower at a farmers market crosses red snapdragons with white snapdragons. ~~The grower has been in business for 20 years and sells mostly to wedding planners.~~ All the offspring are pink. Which inheritance pattern does this show?

Target word: name the pattern **Decides:** The heterozygote is a blend of the two parents.

A	Adjacent category	One color would hide the other; pink would not appear.
B	Correct	A pink heterozygote is intermediate between red and white.
C	Adjacent category	Nothing in the stem links the trait to sex.
D	Confusion set	Codominance would show red and white patches, not a blend.

10. Answer A Concept 7

A pedigree used in class shows a mother with normal color vision whose father was color-blind; she marries a man with normal vision. Red-green color blindness is X-linked recessive. ~~The family lives on a farm and has three dogs.~~ What is the chance that a son of this couple will be color-blind?

Target word: probability for sons **Decides:** The mother is a carrier; sons get their X from her.

A	Correct	Carrier mother passes her affected X to half her sons.
B	Absolute language	She carries one affected X, not two; only half of sons inherit it.
C	Inverted operation	Sons get their X from the mother, not the father.
D	Dropped element	That is the chance for any child, not for a son.

11. Answer B Concept 7 • transfer

A color-blind man and a woman with two normal X alleles have children. What is expected?

Target word: offspring of affected father **Decides:** Father passes X to all daughters, Y to all sons.

A	Inverted operation	Sons receive their X from the mother, who has none affected.
B	Correct	Daughters get his affected X; sons get his Y.
C	Absolute language	Every daughter carries it.
D	Dropped element	Daughters also get a normal X from the mother, so they are carriers.

12. Answer A Concept 8

In a lab using an enzyme from pineapple, students heat samples to 20, 40, 60, and 90 degrees C before adding gelatin. ~~The pineapple was bought on sale and cut by the teacher before class.~~ Activity rises from 20 to 40 degrees C. What result at 90 degrees C best fits enzyme behavior?

Target word: activity at high temperature **Decides:** Far past the optimum the enzyme denatures.

A	Correct	High heat denatures the active site.
B	Confusion set	Enzymes are not used up, but heat still changes their shape.
C	Adjacent category	Enzymes are reusable; the drop comes from denaturing.
D	Absolute language	True only up to the optimum temperature.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Osmosis and tonicity	Water moves across a membrane toward the side with more dissolved solute; cells shrink in hypertonic solutions and swell in hypotonic ones.	Thinking solute moves instead of water, or mixing up which solution is hypertonic.	A cell placed in salt or sugar water; the words hypertonic, hypotonic, isotonic.
2	Organelle structure fits function	A cell's organelles match its job: protein-secreting cells are rich in rough ER and Golgi; energy-hungry cells are packed with mitochondria.	Treating organelles as a memorized list rather than linking them to what the cell does.	A cell type described by its job, asked which organelle would be abundant.
3	Mitosis vs. meiosis	Mitosis makes two identical cells with the parent's chromosome number; meiosis makes four genetically different cells with half the number.	Saying meiosis makes identical cells or that gametes keep the full chromosome number.	Chromosome counts, gametes vs. body cells, growth vs. reproduction.
4	Monohybrid crosses	A cross of two heterozygotes gives a 3:1 phenotype ratio and a 1:2:1 genotype ratio; a 1:1 ratio signals a test cross with a heterozygote.	Mixing up genotype and phenotype ratios.	Two parents that are both Aa, or offspring counts that fall near 3:1 or 1:1.
5	Independent assortment and the product rule	Genes on different chromosomes sort independently, so multiply single-gene probabilities; AaBb x AaBb gives 9:3:3:1.	Adding probabilities for independent genes, or assuming traits travel together.	Two traits in one cross and a question about a combination.
6	Incomplete dominance vs. codominance	In incomplete dominance the heterozygote is a blend; in codominance both alleles show fully and separately.	Calling any heterozygote that looks different from both parents codominant.	A pink offspring from red and white, or spots/patches of both colors.
7	X-linked recessive inheritance	Males have one X, so a recessive X-linked allele shows in any male who has it; a carrier mother passes it to half her sons.	Assuming sons get X-linked alleles from their father.	A trait that shows up mostly in males; a carrier mother.
8	Enzymes and conditions	Enzymes are specific proteins with an optimum temperature and pH; beyond the optimum, the active site changes shape and activity drops.	Thinking higher temperature always speeds an enzyme up, or that enzymes are used up.	A graph with a peak, or temperature/pH raised well past normal body conditions.

Course scope (public description): The Secondary Biology course offers a comprehensive understanding of biological principles and processes, starting from the fundamental cellular structures and functions to the complexities of ecosystem dynamics. Students will delve into key topics such as genetics, evolution, and ecological interactions, gaining insights into the diversity of life and the interconnectedness of biological systems. The course emphasizes critical thinking and problem-solving skills, enabling learners to analyze biological phenomena and apply their knowledge to real-world challenges. Through a blend of theoretical knowledge and practical application, students will engage in hands-on activities, scenarios, and case studies to deepen their understanding. Additionally, the course highlights the impact of environmental changes on ecosystems, fostering an appreciation

for conservation and sustainability. By the end of the course, learners will be well-prepared to address various biological challenges and pursue further studies or careers in the field of biology education.